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## Education

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| 2021/9 - 2026/4 | <b>Doctor of Philosophy</b> , Earth Sciences, Simon Fraser University<br>Thesis: <a href="#">Gaussian process emulation of subglacial drainage for model calibration, uncertainty quantification and efficient coupled ice-sheet modelling</a><br>Supervisors: Dr. Gwenn Flowers, Dr. Derek Bingham |
| 2019/9 - 2021/8 | <b>Master of Mathematics</b> , Applied Mathematics, University of Waterloo<br>Thesis: <a href="#">Mathematical modelling of supraglacial meltwater production and drainage</a><br>Supervisor: Dr. Christine Dow                                                                                     |
| 2014/9 - 2019/4 | <b>Bachelor of Science</b> , Honours Co-operative Mathematical Physics, University of Waterloo                                                                                                                                                                                                      |

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## Employment

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| 2026/8 - present | <b>Research Associate</b> , Center for Climatic Research, University of Wisconsin-Madison                                                                                                                                                                                                                                        |
| 2023/8           | <b>Field Assistant</b> , Geological Survey of Canada <ul style="list-style-type: none"><li>Monitoring alpine hazards, Joffre Lakes, BC</li><li>Install &amp; service timelapse cameras, temperature cables</li></ul>                                                                                                             |
| 2019/5 - 2019/8  | <b>Research Assistant</b> , ECCC Climate Research Division <ul style="list-style-type: none"><li>Estimating power plant CO<sub>2</sub> emissions from satellite data</li><li>Develop &amp; maintain python/bash software for operational analysis</li><li>Research &amp; publish sensor requirements for new satellite</li></ul> |
| 2018/5 - 2018/8  | <b>Undergrad Research Assistant</b> , University of Waterloo Applied Math <ul style="list-style-type: none"><li>Lake ice modelling with MITgcm using high-performance computing clusters</li><li>Develop user manual for future graduate research projects</li></ul>                                                             |
| 2017/9 - 2017/12 | <b>Undergrad Research Assistant</b> , Institute for Quantum Computing                                                                                                                                                                                                                                                            |
| 2016/9 - 2017/4  | <b>Research Assistant</b> , ECCC Climate Research Division                                                                                                                                                                                                                                                                       |

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## Teaching

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| 2025/1 - 2025/4  | <b>Teaching Assistant</b> , Simon Fraser University<br><i>EASC 305: Quantitative Methods for Earth Science</i> <ul style="list-style-type: none"><li>Lead &amp; grade weekly computer lab assignments</li></ul> |
| 2023/1 - 2023-2  | <b>Accessible Learning Lab Aid</b> , Simon Fraser University<br><i>EASC 314: Principles of Glaciology</i> <ul style="list-style-type: none"><li>Individual computer lab support</li></ul>                       |
| 2020/9 - 2020/12 | <b>Teaching Assistant</b> , University of Waterloo<br><i>Math 227: Calculus 3 for Honours Physics</i>                                                                                                           |

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## Mentoring

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2023/4 - 2023/8      **Supervisor**, Orion Villasi, Undergraduate Research Assistant  
*Subglacial hydrology modelling with ISSM*

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## Leadership

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2025/10 - 2026/10      **EX-Officio**, [APECS Canada](#)  
• Ensure continuity and transfer of institutional knowledge

2023/10 - 2025/10      **Vice-Chair Administration & Finance**, [APECS Canada](#)  
• Initiated [pro deal program](#) available to all Canadian polar ECRs  
• Chair of 2024 [Mentor Award](#) committee

2023 - 2025      **Graduate Student Society (GSS) Representative & Secretary**, Department of Earth Sciences Graduate Student Caucus

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## Peer review

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2025      Journal of Geophysical Research: Earth Surface, Nature Communications  
Journal of Glaciology, Journal of Artificial Intelligence Research, IASC cryosphere fellowship program

2024      Nature, Journal of Geophysical Research: Earth Surface,  
Geophysical Research Letters, IASC cryosphere fellowship program

2023      The Cryosphere [**EGU outstanding reviewer award**]  
Geophysical Research Letters

2022      Journal of Glaciology

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## Selected honours & awards

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| 2026/9 - 2028/9 | Canada Postdoctoral Research Award, <i>NSERC Canada</i>         | \$140,000 |
| 2024            | IGS Graham Cogley Award                                         | \$500     |
| 2024/8          | Best oral presentation, <i>IGS model verification symposium</i> | -         |
| 2024/4          | Outstanding Reviewer Award, <i>EGU/The Cryosphere</i>           | -         |
| 2021/9 - 2024/9 | Provost Prize of Distinction, <i>Simon Fraser University</i>    | \$15,000  |
| 2021/9 - 2024/9 | Canada Graduate Scholarship Doctoral, <i>NSERC Canada</i>       | \$105,000 |
| 2021/7          | Outstanding Teaching Assistant Award, <i>UWaterloo</i>          | \$500     |
| 2020/9          | Joseph Wai-Hung Liu Graduate Scholarship, <i>UWaterloo</i>      | \$5,000   |
| 2020/9          | Ontario Graduate Scholarship, <i>Government of Ontario</i>      | \$15,000  |
| 2019/9          | Canadian Graduate Scholarship Masters, <i>NSERC Canada</i>      | \$17,500  |

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## Publications

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- Hill, T.**, Hoffman, M. J., Flowers, G. E., & Bingham, D. (2026). Random forest parameterization of Antarctic subglacial hydrology for coupled ice-flow modelling. *EGUsphere preprint* [submitted 2026-01-29]. doi: [10.5194/egusphere-2026-343](https://doi.org/10.5194/egusphere-2026-343)
- Hill, T.**, Flowers G. E., Bingham, D., & Hoffman, M. J. (2025b) Emulator-based Bayesian calibration of a subglacial drainage model. *Annals of Glaciology*, 66, e22. doi:[10.1017/aog.2025.10016](https://doi.org/10.1017/aog.2025.10016).
- Hill, T.**, Bingham, D., Flowers, G. E., & Hoffman, M. J. (2025a) Computationally efficient subglacial drainage modelling using Gaussian Process emulators: GlaDS-GP v1.0. *Geoscientific Model Development*, 18, 4045-4074. doi: [10.5194/gmd-18-4045-2025](https://doi.org/10.5194/gmd-18-4045-2025)
- Killingbeck, S. F., Rutishauser, A., Unsworth, M. J., Dubnick, A., Criscitiello, A. S., Killingbeck, J., Dow, C. F., **Hill T.**, Booth, Adam D., Main, B., & Brossier, E. (2024) Misidentified subglacial lake beneath the Devon Ice Cap, Canadian Arctic: a new interpretation from seismic and electromagnetic data. *The Cryosphere*, 18(8), 3699-3722. doi: [10.5194/tc-18-3699-2024](https://doi.org/10.5194/tc-18-3699-2024)
- Hill, T.**, Flowers, G. E., Hoffman, M. J., Bingham, D., and Werder, M. A. (2024) Improved representation of laminar and turbulent sheet flow in subglacial drainage models. *Journal of Glaciology*, 70, e24. doi: [10.1017/jog.2023.103](https://doi.org/10.1017/jog.2023.103) [2024 IGS Graham Cogley student paper award]
- Hill, T.** and Dow, C. F. (2023) The impact of surface melt rate and catchment characteristics on Greenland Ice Sheet moulin inputs. *The Cryosphere*, 17, 2607–2624 doi: [10.5194/tc-17-2607-2023](https://doi.org/10.5194/tc-17-2607-2023)
- Hill, T.** and Dow, C. F. (2021) Modeling the Dynamics of Supraglacial Rivers and Distributed Meltwater Flow with the Subaerial Drainage System (SaDS) Model. *Journal of Geophysical Research: Earth Surface*, 126, e2021JF006309. doi: [10.1029/2021JF006309](https://doi.org/10.1029/2021JF006309)
- Hill, T.**, Dow, C. F., Bash, E. A., and Copland, L. (2021) Application of an improved surface energy balance model to two large valley glaciers in the St. Elias Mountains, Yukon. *Journal of Glaciology*, 67(262), 297-312. doi: [10.1017/jog.2020.106](https://doi.org/10.1017/jog.2020.106)
- Nassar, R., Mastrogiacomio, J. P., Bateman-Hemphill, W., McCracken, C., MacDonald, C. G., **Hill, T.**, O'Dell, C., Kiel, M., & Crisp, D. (2021). Advances in quantifying power plant CO<sub>2</sub> emissions with OCO-2. *Remote Sensing of Environment*, 264, 112579. doi: [10.1016/j.rse.2021.112579](https://doi.org/10.1016/j.rse.2021.112579)
- Hill, T.**, and Nassar, R. (2019) Pixel size and revisit rate requirements for monitoring power plant CO<sub>2</sub> emissions from space. *Remote Sensing*, 11(13): 1608. doi: [10.3390/rs11131608](https://doi.org/10.3390/rs11131608)
- Nassar, R., **Hill, T. G.**, McLinden, C., Wunch, D., Jones, D., and Crisp, D. (2017) Quantifying CO<sub>2</sub> emissions from individual power plants from space. *Geophysical Research Letters*, 44(19), 10045-10053. doi: [10.1002/2017GL074702](https://doi.org/10.1002/2017GL074702)

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## Selected conferences & presentations

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- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2026) Subglacial drainage modelling: from fundamental physics to machine-learning emulation. IGS global seminar series, Online. [Invited]
- Hill, T.**, Flowers, G., Hoffman, M., and Bingham, D. (2025) Emulator-based subglacial drainage modelling. Online seminar, Mathematics on Ice forum.

- Hill, T.**, Flowers, G., Hoffman, M., and Bingham, D. (2025) Random-Forest Parameterization of Antarctic Subglacial Hydrology. Poster presentation at the 2025 WAIS Workshop, Coupeville, USA.
- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2025) Subglacial drainage modelling and Bayesian calibration using Gaussian Process emulators. Oral presentation at the 2025 EGU General Assembly, Vienna, Austria, EGU25-3771. <https://doi.org/10.5194/egusphere-egu25-3771>  
[Invited]
- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2025) Subglacial drainage modelling and Bayesian calibration using Gaussian Process emulators. Seminar at SFU Earth Sciences Research Day, Vancouver, Canada.
- Hill, T.**, Flowers, G., Bingham, D., and Hoffman, M. (2024) Gaussian process emulation of subglacial drainage. Oral presentation at the International Symposium on Verification and Validation of Cryospheric Models, Newcastle Upon Tyne, UK. **[Best Oral Presentation Award]**
- Hill, T.**, Flowers, G., Bingham, D., Hoffman, M., and Werder, M. (2023) Improved representation of laminar and turbulent sheet flow in subglacial drainage models. Oral presentation at the AGU Fall Meeting, San Francisco, USA, C51B-08.
- Hill, T.**, Flowers, G., Bingham, D., Hoffman, M., and Werder, M. (2023) Laminar and turbulent subglacial water flow. Oral presentation at the Annual Meeting of the Northwest Glaciologists, Seattle, USA.
- Killingbeck, S., Killingbeck, J., **Hill, T.**, Main, B., Brossier, E., Unsworth, M., Dow, C., Dubnick, A., Criscitiello, A., and Rutishauser, A. (2022) Combined Geophysical Techniques Constrain the Devon Ice Cap, Canadian High Arctic, Subglacial Environment. Oral presentation at AGU Fall Meeting, Chicago, USA, C42B-05.
- Hill, T.**, Flowers, G., and Bingham, D. (2022) Gaussian process emulation of subglacial drainage. Oral presentation at Annual Meeting of the Northwest Glaciologists, Moscow, USA.
- Siu, K., Dow, C. F., Morlighem, M., McCormack, F., and **Hill, T.** (2022) Modelling Subglacial Hydrology under Future Climate Scenarios in Wilkes Subglacial Basin, Antarctica. Oral presentation at EGU General Assembly 2022, Vienna, Austria, EGU22-424.
- Hill, T.**, and Dow, C. F. (2021) Modeling the Dynamics of Supraglacial Rivers and Distributed Meltwater Flow on the Greenland Ice Sheet With the Subaerial Drainage System (SaDS) Model. Poster presentation at AGU Fall Meeting, New Orleans, USA.
- Hill, T.**, and Dow, C. F. (2021) Modelling glacier melt rates and surface hydrology at the basin scale. Poster presentation at CGU Student Conference, Online.